Uses of weighted arithmetic mean

Weighted arithmetic mean is used in:

- The construction of index numbers.
- Comparison of results of two or more groups where number of items in the groups differs.
- Computation of standardized death and birth rates.

Example 5.5

The weights assigned to different components in an examination or Component Weightage Marks scored

Component	Weightage	Marks scored
Theory	4	60
Practical	3	80
Assignment	1	90
Project	2	75
	10	

Calculate the weighted average score of the student who scored marks as given in the table

Solution:

Component	Marks scored (x_i)	Weightage (w_i)	$w_i x_i$
Theory	60	4	240
Practical	80	3	240
Assignment	90	1	90
Project	75	2	150
Total		10	720

Weighted average, $\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$

$= 720/10$

$= 72$



NOTE

If the weights are not assigned, then the mean is

$= (60 + 80 + 90 + 75)/4$

$= 76.25$

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